

Parcel Reality Bench Start Here

How to run it, in plain steps, on an M1 Mac. Three paths, from zero setup to fully automatic. Pick the first one you have time for. Every command is copy-paste ready.

The 60-second version

Parcel Reality Bench (prbench) is a test that grades AI models on messy, real-format county property records from the Indiana–Michigan border. Each test item is a property plus six comparable sales. Hidden inside are traps real appraisers hit every week: prices stored in a state file format, family sales at half price, one sale split across three rows, duplicate records, comps from the wrong state, and properties whose mailing address points to the wrong county.

A 200-line deterministic records pipeline scores 94.5 on the hard tier. Taking the data at face value scores 45.7. Where frontier AI models land between those two numbers is the story.

PATH	WHAT YOU NEED	TIME	WHAT YOU GET
1 Browser	Nothing. Your Claude login.	10–30 min	A real Claude score, receipts, a downloadable result file
2 Your Mac	Terminal + API keys	30 min	Pinned model scores (Claude, GPT, Gemini) + a leaderboard site
3 Automatic	GitHub + Cloudflare	30 min once	Runs every morning, benchmarks new models the day they ship, publishes to bench.neuruh.com

WHY THE README LOOKED BROKEN

README.md is written in Markdown, a format meant for GitHub and code editors. TextEdit shows the raw symbols instead of formatting it. This PDF replaces it for reading. The README stays in the folder because GitHub will render it properly once the repo is pushed.

Path 1 — Run it in your browser

No Terminal. No keys. This runs Claude through your Claude account and scores it on the page.

- 1. Open <https://claude.ai/artifact/KdfUhqRUtjrjuFSy2M5Y2N>
- 2. Set Items to 30. Leave Claude tier on Default.
- 3. Press Run benchmark. Approve the one-time permission prompt. It uses your Claude usage.
- 4. Leave the tab open. Each item takes 20–60 seconds; 30 items is roughly 15–25 minutes.
- 5. When it says Done, press Download results (.json). It lands in your Downloads folder.
- 6. Drop that file into a Claude chat. It gets verified and filed into the main ledger.

YOU WILL SEE	WHAT IT MEANS
Big brass number	Claude’s composite score, 0–100, next to the two baselines
Failure map	Per trap type: held (80%+), partial (40–79%), failed (under 40%)
Ledger head	A fingerprint of every answer. Change one character anywhere and it changes.
Rate limited	Normal on long runs. Wait a minute, press Resume. Nothing is lost.

LIMIT OF THIS PATH

This measures “Claude on claude.ai at the Default tier,” not a named model such as claude-opus-5. Good for knowing where you stand. Not good enough for a public claim. Public numbers come from Path 2 or 3, where the exact model ID is recorded.

Path 2 — Run it on your Mac

Everything below happens in Terminal (Cmd-Space, type Terminal, Enter). Paste one block at a time and compare against “expect” before moving on.

Step 1. Put the project in its home folder

Moves the zip contents to ~/neuruh/prbench (your home folder → neuruh → prbench) and steps into it. Every later command runs from there.

```
cd ~/Downloads
unzip -o prbench.zip -d ~/neuruh
cd ~/neuruh/prbench
python3 --version
```

Expect: a line like Python 3.9.6 or higher. No installs needed; prbench uses only what ships with Python.

Step 2. Create a secret salt, stored in macOS Keychain

The salt seals the private holdout answers so you can later prove you never changed them. Keychain keeps it off disk and out of screenshots.

```
security add-generic-password -a "$USER" -s prbench-salt -w "$(openssl rand -hex 32)"
security find-generic-password -a "$USER" -s prbench-salt -w | cut -c1-8
```

Expect: 8 random characters. That is the start of your salt; the rest stays hidden.

Step 3. Generate the test items and seal the holdouts

```
python3 -m prbench generate
python3 -m prbench commit --salt "$(security find-generic-password -a "$USER" -s prbench-salt -w)"
```

Expect: four lines starting [public], [holdout], [public_hard], [holdout_hard], then commitment hashes. Those hashes are what you publish before any model runs.

Step 4. Store your API keys in Keychain

Get keys from: Anthropic → console.anthropic.com → API Keys. OpenAI → platform.openai.com/api-keys. Google → aistudio.google.com/apikey. Each command below prompts for the key; paste it and press Enter. Nothing shows while you paste. That is normal.

```
security add-generic-password -a "$USER" -s anthropic-api-key -w
security add-generic-password -a "$USER" -s openai-api-key -w
security add-generic-password -a "$USER" -s gemini-api-key -w
```

Skip any provider you do not have yet. prbench logs skipped (no API key) and moves on.

Step 5. Load the keys into this Terminal window and run

--bootstrap marks every older model as already seen so the first real run only benchmarks the current flagships pinned in bench.config.json.

```
export ANTHROPIC_API_KEY="$(security find-generic-password -a "$USER" -s anthropic-api-key -w)"
export OPENAI_API_KEY="$(security find-generic-password -a "$USER" -s openai-api-key -w 2>/dev/null)"
export GEMINI_API_KEY="$(security find-generic-password -a "$USER" -s gemini-api-key -w 2>/dev/null)"
python3 -m prbench auto --bootstrap
python3 -m prbench auto
```

Expect: a model list, then progress per split, then PASS — chain intact. Cost is billed by each provider; budget roughly \$5–\$30 per model for all four splits (estimate; depends on model price and answer length).

Step 6. Look at the leaderboard

```
open site/index.html
```

Opens in your browser. Hard tier is the default page; tabs switch to Standard and Holdout.

BEFORE RUNNING, ADD THE OPENAI AND GEMINI MODEL IDS

Open bench.config.json in TextEdit (Finder → ~/neuruh/prbench → right-click → Open With → TextEdit). Find “pinned_models”. Add the current flagship IDs from each provider’s models page. Save. Only Anthropic IDs are filled in today.

Path 3 — Automatic, live under Neuruh

Goal: a private GitHub repo that runs the bench every morning at 9:17 Eastern, benchmarks any new model the day it ships, verifies the whole ledger, and publishes to bench.neuruh.com. neuruh.com is already on Cloudflare, so the site goes on Cloudflare Pages alongside the rest of the estate.

Hand this to Claude Code on the M1

Open Terminal, run `cd ~/neuruh/prbench && claude`, paste the block below. It stops and asks for anything it cannot find itself.

```
Read the Notion page "CONVERGENCE DELTA – Parcel Reality Bench (prbench)" under the NEURUH TOPOLOGY NOTEBOOK first. Do not re-run work it marks done.

Then in ~/neuruh/prbench:
1. Run: python3 -m prbench verify. Stop and report if it does not print PASS.
2. Run git status. Confirm data/holdout/key.jsonl and data/holdout_hard/key.jsonl are gitignored and never staged. Never commit them, never print them.
3. brew install gh if missing; gh auth login if needed. Create a PRIVATE repo named prbench with gh repo create, push main.
4. Set repo secrets with gh secret set, reading values from macOS Keychain:
  ANTHROPIC_API_KEY <- anthropic-api-key
  OPENAI_API_KEY    <- openai-api-key
  GEMINI_API_KEY    <- gemini-api-key
  HOLDOUT_KEYS_B64  <- tar czf - data/holdout/key.jsonl data/holdout_hard/key.jsonl | base64
  CLOUDFLARE_API_TOKEN, CLOUDFLARE_ACCOUNT_ID <- ask me; do not guess.
5. Create Cloudflare Pages project "prbench" with wrangler and attach custom domain bench.neuruh.com (neuruh.com zone is on Cloudflare).
6. gh workflow run bench.yml, then gh run watch. Report site URL and ledger head.
Show every command and its output. Never echo secret values.
```

What the daily loop does

STEP	ACTION	SAFETY
Discover	Asks Anthropic, OpenAI, Google for their model lists; filters to real chat models	Regex filters in config, not code
Benchmark	Runs up to 4 new models per day on all splits	Concurrency lock: one ledger writer
Verify	Recomputes every hash and re-scores every stored answer	Site is not rebuilt if anything fails
Publish	Builds and deploys the leaderboard	Holdout answer keys deleted before commit or deploy
Draft	Writes a post draft per new model with numbers filled in	Nothing auto-posts
Saturation	Flags any trap every model beats	Tells you what to harden next

Where everything lives

FOLDER OR FILE	WHAT IT HOLDS	PUBLIC?
prbench/	The program (generator, scorer, receipts, leaderboard, CLI)	Yes
bench.config.json	Which models to test, which splits, filters	Yes
data/public*/	Test items and answer keys anyone can check	Yes
data/holdout*/key.jsonl	Sealed answers. The whole point of the holdout.	NEVER
results/	One file per model run, every raw answer included	Yes
receipts/ledger.jsonl	The hash chain. Append-only.	Yes
pilot/	The first claude.ai run and its analysis	Yes
site/	The built leaderboard pages	Yes
drafts/	Auto-written post drafts and saturation report	Private until you post

If something breaks

YOU SEE	DO THIS
command not found: python3	Run xcode-select --install, accept the prompt, retry.
No such file or directory	You are in the wrong folder. Run cd ~/neuruh/prbench first.
401 / invalid x-api-key	Key typo. Delete and re-add: security delete-generic-password -a "\$USER" -s anthropic-api-key, then Step 4.
FAIL – chain broken	Something edited a result or receipt. Do not publish. Paste the output into Claude.
429 / rate limit	Provider throttling. Re-run python3 -m prbench auto later; finished work is kept.